

1.3 Meet your project: the Cold Chain Log

i You're going to spend the next four weeks building one thing. Everything on this board is a step toward it.

Before you write any code, focus on understanding what you're building — and why a pharmacy would care. That's exactly how we'd want you to approach a real feature.

The project

A **Cold Chain Log**: a digital record that helps pharmacies monitor fridge temperatures for vaccines, insulin, and other medicines.

- The fridge must stay between **2°C and 8°C**.
- Temperatures need to be recorded every day.
- If the fridge goes out of range, the stock may not be safe for patients and someone needs to know quickly.

In short: today this is often a paper chart taped to the fridge. You're building the digital version.

Why does a temperature excursion matter so much to a pharmacy? What's the actual consequence?

A temperature excursion matters to a pharmacy because it can silently ruin the chemical and biological stability of medications. The actual consequence is that active ingredients degrade, lose potency, or form harmful byproducts, meaning a patient receives a useless dose of a life-saving drug or vaccine while believing they are fully protected.

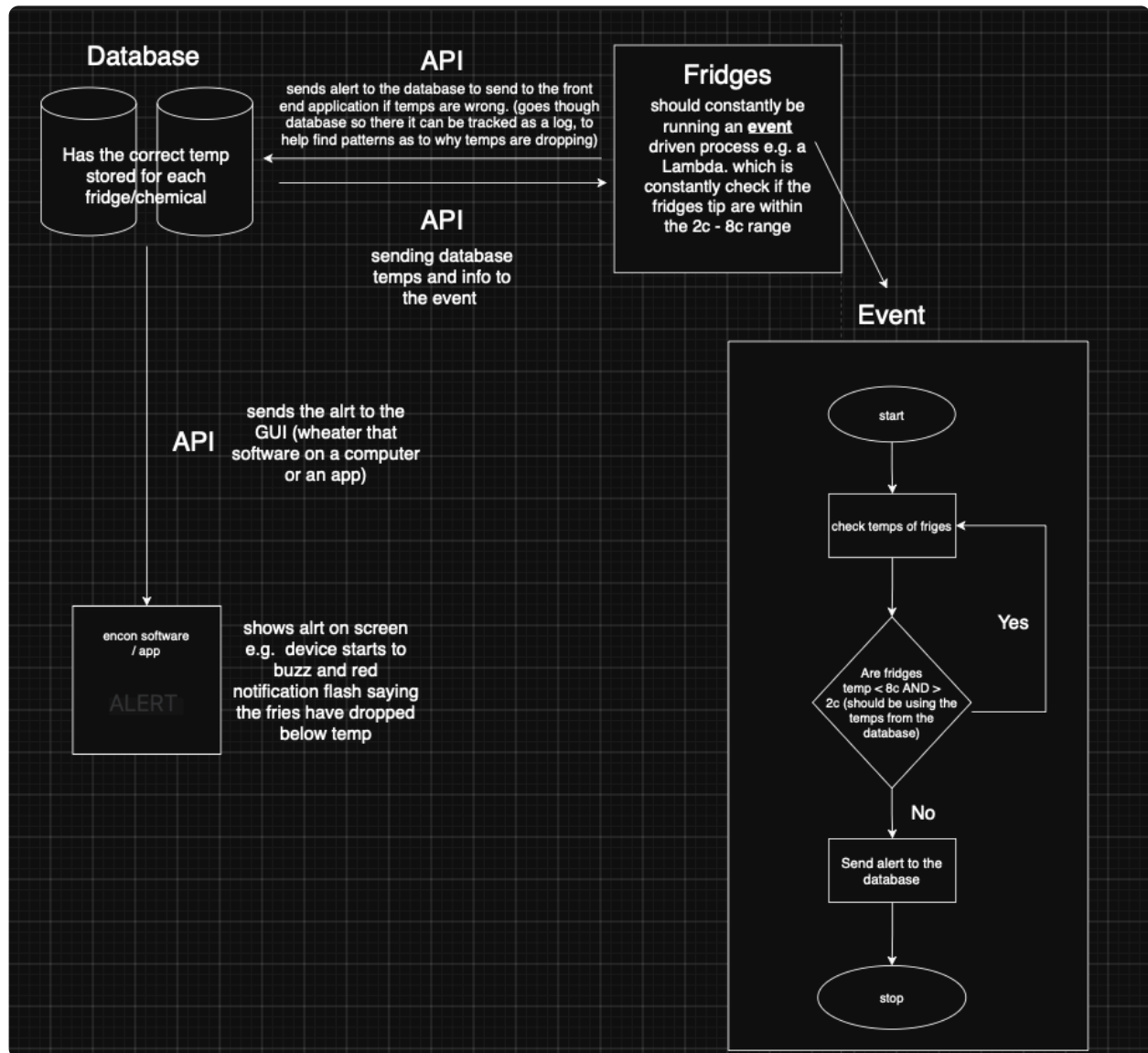
Why should the server decide whether a reading is an excursion, rather than the app that sends it?

From what I have gathered, the server should decide which temperature excursion threshold to use because it is more reliable than having a person look at an app and manually input it. I'm not entirely sure.

Why does an alert need to happen on its own, rather than someone noticing a red row in a table?

An alert will be more effective for keeping the temperature within the correct range because when the fridge temperature drops below or rises above the necessary threshold, an alert will instantly inform a staff member so they can act quickly and avoid major damage to the medications. However, if it were just a red row in a table, no one would be aware of the temperature change until someone noticed it. This delay could range from minutes to days or even weeks, putting the medical products at risk of becoming damaged or obsolete.

Diagram of model



I also believe there are certain functions that need to be implemented, but that is it. For example, there may be five functions that need to be made, and I need to stick to those five. No more and no less. I have identified potentially three of the five.

1. Send the alert automatically whenever the temperatures are out of range (the most obvious one).
2. Pharmacists being able to log temperatures.
3. Any pharmacist or person in the company should be able to view an item and see everything about it (limited access control, maybe).

My other idea is that, because it is a made-up company, the reads and logs from my project progress need to be accessible to Hamza and other people, so it might be going down that angle.

I am not sure yet.

 Question / Queries / Push back

- **Server deciding whether a reading is an excursion** — this statement gave me **pushback**. I was confused about what it meant and still don't have a proper grasp of it as of this moment (4 Aug, 13:44).
- This is one question I am confused about. I believe it is linked to the project, but I am not sure if the answer I gave was enough to resolve the question being asked.